

GLM-5V-Turbo - Overview - Z.AI DEVELOPER DOCUMENT

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[Skip to main content](#)

[Overview - Z.AI DEVELOPER DOCUMENT home page](#)

[English](#)

[Search...](#)

[⌕](#)

[API Keys](#)

[Payment Method](#)

[Guides](#)

[API Reference](#)

[Scenario Example](#)

[Coding Plan](#)

[Released Notes](#)

[Terms and Policy](#)

[Help Center](#)

[Get Started](#)

[Quick Start](#)

[Overview](#)

[Pricing](#)

[Core Parameters](#)

[SDKs Guide](#)

[Migrate to GLM-5](#)

[Language Models](#)

[GLM-5](#)

[GLM-5-Turbo](#)

[GLM-4.7](#)

[GLM-4.6](#)

[GLM-4.5](#)

[GLM-4-32B-0414-128K](#)

[Vision Language Models](#)

[GLM-5V-Turbo](#)

[GLM-4.6V](#)

[GLM-OCR](#)

[AutoGLM-Phone-Multilingual](#)

[GLM-4.5V](#)

[Image Generation Models](#)

[GLM-Image](#)

[CogView-4](#)

[Video Generation Models](#)

[CogVideoX-3](#)

[Vidu Q1](#)

[Vidu 2](#)

[Image Generation Models](#)

[CogView-4](#)

[Audio Models](#)

[GLM-ASR-2512](#)

[Capabilities](#)

[Thinking Mode](#)

[Deep Thinking](#)

[Streaming Messages](#)

Tool Streaming Output
Function Calling
Context Caching
Structured Output
Tools
Web Search
Stream Tool Call
Agents
GLM Slide/Poster Agent(beta)
Translation Agent
Video Effect Template Agent
On this page
Overview
Capability
Usage
Resources
Introducing GLM-5V-Turbo
Multimodal Coding Foundation
Systematic Upgrades Across Four Layers
Official Skills
Examples
Quick Start
Vision Language Models
GLM-5V-Turbo
Copy page

Overview

GLM-5V-Turbo is Z.AI's first multimodal coding foundation model, built for vision-based coding tasks. It can natively process multimodal inputs such as images, video, and text, while also excelling at long-horizon planning, complex coding, and action execution. Deeply optimized for agent workflows, it works seamlessly with agents such as Claude Code and OpenClaw to complete the full loop of "understand the environment → plan actions → execute tasks".

Positioning

Multimodal Coding Model

Input Modality

Video / Image / Text / File

Output Modality

Text

Context Length

200K

Maximum Output Tokens

128K

Capability

Thinking Mode

Offering multiple thinking modes for different scenarios

Vision Comprehension

Powerful vision understanding capabilities, with support for images, video, and files

Streaming Output

Support real-time streaming responses to enhance user interaction experience

Function Call

Powerful tool invocation capabilities, enabling integration with various external toolsets

Context Caching

Intelligent caching mechanism to optimize performance in long conversations

Usage

Frontend Recreation

GUI Autonomous Exploration and Recreation

Code Debugging

OpenClaw

Resources

API Documentation: Learn how to call the API.

Introducing GLM-5V-Turbo

1

Multimodal Coding Foundation

Across benchmarks for multimodal coding and agentic tasks, as well as pure-text coding, GLM-5V-Turbo delivers strong performance with a smaller model size.

Multimodal Coding and Agentic Tasks

Pure-Text Coding Tasks

In multimodal coding and agentic tasks, GLM-5V-Turbo achieved leading results across benchmarks for design-to-code generation, visual code generation, multimodal retrieval and question answering, and visual exploration. It also delivered strong performance on benchmarks such as AndroidWorld and WebVoyager, which evaluate an agent's ability to operate in real GUI environments.

2

Systematic Upgrades Across Four Layers

GLM-5V-Turbo combines strong vision and coding capabilities while achieving leading performance with a smaller parameter size, powered by systematic upgrades across model architecture, training methods, data construction, and tooling:

Native Multimodal Fusion: From pretraining through post-training, the model continuously strengthens visual-text alignment. Combined with the new CogViT vision encoder and an inference-friendly MTP architecture, it improves multimodal understanding and reasoning efficiency.

30+ Task Joint Reinforcement Learning: During RL, the model is jointly optimized across 30+ task types, spanning STEM, grounding, video, GUI agents, and coding agents, resulting in more robust gains in perception, reasoning, and agentic execution.

Agentic Data and Task Construction: To address the scarcity of agent data and the difficulty of verification, we built a multi-level, controllable, and verifiable data system, and injected agentic meta-capabilities during pretraining to strengthen action prediction and execution.

Expanded Multimodal Toolchain: Adds multimodal tools such as box drawing, screenshots, and webpage reading (including image understanding), extending agent capabilities from pure text to visual interaction and supporting a more complete perception-planning-execution loop.

Official Skills

Beyond vision-based coding and Claw-style tasks, GLM-5V-Turbo also shows major gains in a broader range of agentic scenarios, including multimodal search, deep research, GUI agents, and perceptual grounding. To support these use cases, we provide a set of official Skills.

Image Captioning

Visual Grounding

Document-Grounded Writing

Resume Screening

Prompt Generation

Additionally, based on the previously released specialized models GLM-OCR and GLM-Image, we have created five Skills to support a wider range of scenarios and tasks.

The above Skills are now available on ClawHub, Install Now!

Examples

Web Page Coding

Website Generation

Document Comprehension & Writing

Video Object Tracking

Input

Please recreate the mobile pages based on the design mockups in the images. The left side shows the welcome page, and the center shows the homepage image. You will also need to create mockups for the remaining two pages.

Output

Quick Start

cURL

Python

Java

Basic Call

```
curl -X POST \
  https://api.z.ai/api/paas/v4/chat/completions \
  -H "Authorization: Bearer your-api-key" \
  -H "Content-Type: application/json" \
  -d '{
    "model": "glm-5v-turbo",
    "messages": [
      {
        "role": "user",
        "content": [
          {
            "type": "image_url",
            "image_url": {
              "url": "https://cloudcovert-1305175928.cos.ap-
guangzhou.myqcloud.com/%E5%9B%BE%E7%89%87grounding.PNG"
            }
          },
          {
            "type": "text",
            "text": "Where is the second bottle of beer from the right on the
table? Provide coordinates in [[xmin,ymin,xmax,ymax]] format"
          }
        ]
      }
    ],
    "thinking": {
      "type": "enabled"
    }
  }'
```

```
}'
```

Streaming Call

```
curl -X POST \
  https://api.z.ai/api/paas/v4/chat/completions \
  -H "Authorization: Bearer your-api-key" \
  -H "Content-Type: application/json" \
  -d '{
    "model": "glm-5v-turbo",
    "messages": [
      {
        "role": "user",
        "content": [
          {
            "type": "image_url",
            "image_url": {
              "url": "https://cloudcovert-1305175928.cos.ap-
guangzhou.myqcloud.com/%E5%9B%BE%E7%89%87grounding.PNG"
            }
          },
          {
            "type": "text",
            "text": "Where is the second bottle of beer from the right on the
table? Provide coordinates in [[xmin,ymin,xmax,ymax]] format"
          }
        ]
      }
    ],
    "thinking": {
      "type": "enabled"
    },
    "stream": true
  }'
```

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Yes

No

GLM-4-32B-0414-128K

GLM-4.6V

x

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